

0914 U1 investigate questions

1. A city's annual car auction sells vintageolder vehicles. A car collector was curious to know whether the mean selling price of the vehicles was different between the auction years 2024 and 2026. Approximately 2,000 vehicles were sold at each auction, and none of the vehicles were sold at both auctions. The collector took a random sample of 36 vehicles sold at the 2024 auction and calculated a mean selling price of \$106,400 and a standard deviation of \$32,000. The collector also took a random sample of 42 vehicles sold at the 2026 auction and calculated a mean selling price of \$96,500 and a standard deviation of \$22,000.

Complete the appropriate inference procedure at the 98% confidence level to estimate the difference between the population means of the vehicle selling prices for 2024 and 2026.

Respond to parts A, B, C, D, and E.

Part B

Identify the type of study being conducted.

A country's transportation department wants to conduct a survey to collect data from adults, ages 18 and older, in that country. The survey will be focused on commuting to work, including commute time in minutes, method of transportation used, and other information.

Country A is divided into many regions, and each region will be asked to send out a survey and collect the data. The transportation department will allow each region to use any method it prefers to survey the adults.

Three different regions will use the following sampling methods:

Region X: Each adult who lives in the region will receive a survey and will be required to complete the survey by mail. Adults who do not complete the survey will be sent reminders to do so until they have completed the survey.

Region Y: The region is divided into six areas that are similar to each other. Two of the six areas will be randomly selected, and each adult who lives in those two areas will receive a survey and be asked to complete the survey by mail. The table gives the results of the data collection for the two areas in Region Y for the question "What is your average daily commute time to work in minutes?"

Region Z: Each adult who lives in the region and has responded to a previous official survey will receive a survey and be asked to complete it by mail.

Summary Statistics

Area	$\bar{x}$	n	Min	Q1	Median (Q2)	Q3	Max
One	55	2,887	7	35	53	74	101
Five	42	4,502	5	24	32	68	83

2. Respond to parts A, B, C, D, and E, and any subparts.

Part E

Manuel wants to perform a study about the average daily commute time to work for Area Three and Area Six in Region Y. He believes there is a difference in the commute times for the two areas. **Determine** a valid investigative question for Manuel's study. Ensure all three components of a valid investigative question are included in your response.

3. Respond to parts A, B, C, D, and E, and any subparts.

Part B

i. **Explain** why the results of the sampling method for Region Y can be generalized to the population of the region.

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- ii. Tom, a member of the planning committee for Region Z, proposed Region Z's sampling method. Tom claims the sampling method for Region Z may lead to a higher survey response rate than the sampling methods used for the other two regions and could better represent the entire population in Region Z. **Explain** why Tom's claim is not correct.

4. Andrea, a cybersecurity specialist at a bank, was concerned about the security of the bank's website. She was asked to compare two encryption processes where information is scrambled and cannot be easily read methods that could be used to protect the website's banking information. Her department was concerned that the older encryption version, DES, may be weaker than the newer encryption version, AES. One measure of how well an encryption method works is session duration, how long the information is accessed in seconds. A very long session duration may indicate that there has been unauthorized access to the data, which means that the encryption method needs to be more secure. Her department wanted her to study the two encryption methods.

Therefore, Andrea conducted a study of the two types of encryption methods. Andrea randomly collected this information from bank customers who used the website over the previous six months. For each session of the sampled customers, Andrea collected the session duration, in seconds, and the type of encryption method used. The table displays the summary statistics for her study.

Summary Statistics

Encryption Method	n	$\bar{x}$	s	Min	Q1	Median (Q2)	Q3	Max
DES	2,865	782	768	1	226	543	1,107	6,950
AES	4,706	794	803	1	233	794	1,091	7,190

Respond to parts A, B, C, D, and E, and any subparts.

Part E

Determine the population for a valid investigative question for Andrea's study.

5. A city's public health department recently started a "Hydration and Health" campaign to encourage adults to drink enough water each day. In the city, doctors have reported that many adults substitute caffeinated beverages, such as coffee and tea, for water. They are worried this might keep adults from staying properly hydrated.

To look into this issue, the department surveyed a random sample of 300 adults from across the city. Each person filled out a short survey about their beverage preference and whether they meet the daily water intake recommendation. The following variables are used to create a table and mosaic plot.

- Beverage Preference: Coffee or Tea
- Meets Water Intake Recommendation: Yes or No

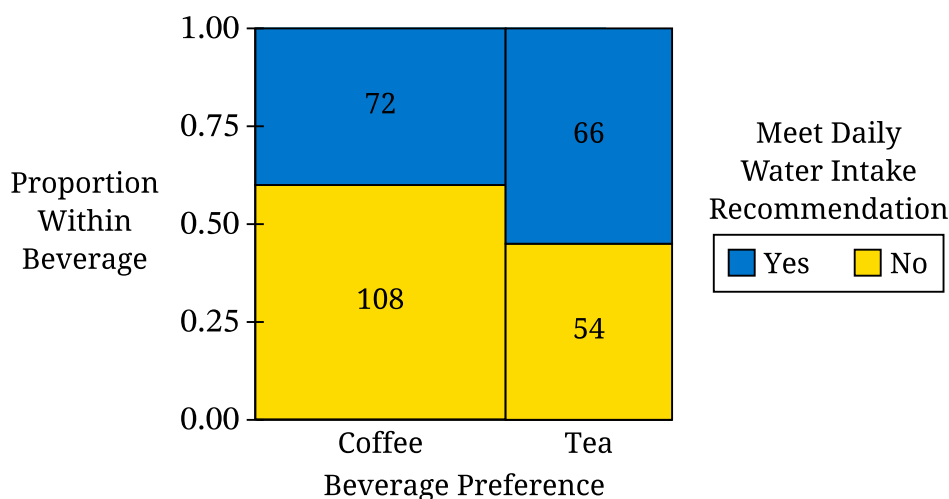
The department plans to use the survey results to help create educational materials for the campaign. City officials also hope the information will show whether new ideas—like adding free water refill stations—might help adults who are not drinking enough water. The survey results from the 300 residents are shown in the table.

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Beverage Preference and Meets Water Intake Recommendation

		Meet Daily Water Intake Recommendation		
		Yes	No	Total
Beverage Preference	Coffee	72	108	180
	Tea	66	54	120
	Total	138	162	300

Beverage Preference and Meets Water Intake Recommendation



Respond to parts A, B, C, D, and E, and any subparts.

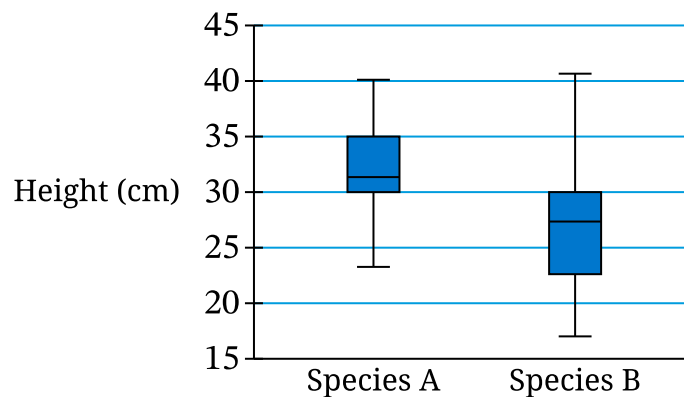
Part E

The public health department is placing free water refill stations throughout the city to study whether doing so increases the proportion of adults across the city who meet the daily water intake recommendation. **Determine** a valid investigative question for the public health department's study. Ensure all three components of a valid investigative question are included in your response.

6. A team of biologists is investigating how two plant species commonly used in ecological restoration process of helping damaged or destroyed ecosystems regain their natural structure and function—Species A, a native grass, and Species B, a wildflower—respond to limited water availability. Although both species typically grow at the same rate under similar soil conditions, their growth may differ when water is limited. To study this possibility, the biologists randomly assigned Species A seeds to 35 plots and Species B seeds to 35 plots that were located in a controlled greenhouse environment with restricted watering. After 12 weeks, they recorded the height, in centimeters (cm), of every plant. The data is displayed for the resulting height distributions of both species.

Plant Heights for Two Species

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Respond to parts A, B, C, D, and E, and any subparts.

Part E

The team of biologists plans to conduct a larger follow-up study to determine whether Species A plants grow taller when they receive a limited amount of water than when they receive a normal amount of water. **Determine** the alternative hypothesis the team should use for their investigative question.